



AI Workforce

Autonomous Business Workflow Automation Platform

ENTERPRISE REQUIREMENTS SPECIFICATION

Business Requirements Document

Complete production-grade business and product requirements for the AI Workforce autonomous workflow automation platform.

DOCUMENT ID

BRD-AIWF-2026-001

VERSION

v1.1.0

DATE

October 2026

DOCUMENT CONTROL

Field	Details
Document Title	Business Requirements Document — AI Workforce Platform
Document ID	BRD-AIWF-2026-001
Version	1.0.0
Status	Draft — Pending Stakeholder Review
Classification	Confidential — Internal Use Only
Created Date	2026-10-07
Last Updated	2026-10-07
Prepared By	Product Architecture Team
Reviewed By	Pending
Approved By	Pending
Next Review Date	2026-11-07

Revision History

Version	Date	Author	Change Summary
0.1	2026-10-07	Product Architecture Team	Initial Draft
1.0	2026-10-07	Product Architecture Team	Complete BRD — First Release for Review

Document Purpose

This Business Requirements Document (BRD) defines the complete business and product requirements for the **AI Workforce** platform. It serves as the authoritative source of truth for all downstream architectural, engineering, and product decisions. This document does not contain implementation details, infrastructure decisions, or code. Those will be addressed in separate Architecture Design Documents (ADDs) and Technical Specification Documents (TSDs) in subsequent phases.

Table of Contents

Executive Summary

Product Vision & Mission

Problem Statement

Business Opportunity

Target Users & Organizations

User Personas

Business Goals & Product Goals

Success Criteria & KPIs

Scope Definition

Assumptions, Constraints & Dependencies

User Roles & Permission Concepts

Multi-Tenant Architecture Requirements

Functional Requirements

13.1 Identity & Access Management

13.2 Organization & Workspace Management

13.3 AI Agent Management

13.4 Knowledge Management & RAG

13.5 Tools & Integrations

13.6 Workflow Automation

13.7 Agentic Orchestration

13.8 Human-in-the-Loop (HITL)

13.9 Guardrails & Governance

13.10 Execution & Monitoring

13.11 AI Evaluation

13.12 Observability

13.13 Analytics & Reporting

13.14 Administration

13.15 Developer / API Platform

• Non-Functional Requirements

• Business Rules

• Security Requirements

• Privacy Requirements

• Audit Requirements

• Reliability & Fault Tolerance Requirements
• Scalability & Performance Requirements
• Data Lifecycle Requirements
• AI Model Management Requirements
• Lifecycle Definitions 23.1 Workflow Lifecycle
• 23.2 Agent Lifecycle
• 23.3 Knowledge Lifecycle
• 23.4 User Lifecycle
• 23.5 Organization Lifecycle
• Risk Management
• Compliance Considerations
• Real-World Business Scenarios
• Future Extensibility
• Product Roadmap MVP Scope
• Phase 2 Scope
• Phase 3 Scope
• Future Vision
• Acceptance Criteria Framework
• Risks & Mitigations
• Open Questions & Decisions Required
• Final Product Requirement Summary

1. EXECUTIVE SUMMARY

1.1 What is AI Workforce?

AI Workforce is a multi-tenant, enterprise-grade SaaS platform that enables organizations to design, deploy, and operate a fleet of AI agents as an autonomous digital workforce. The platform allows businesses to automate complex, multi-step business processes by combining AI reasoning, organizational knowledge (RAG), external tool integrations, visual workflow orchestration, and human-in-the-loop oversight — all governed by security, audit, and compliance controls.

This is not a chatbot product. This is not a RAG demo. This is not a single-use-case AI automation tool.

AI Workforce is a horizontal business automation operating system built for enterprise-grade adoption across industries, use cases, and organization sizes.

1.2 Strategic Positioning

AI Workforce sits at the intersection of:

- **AI Agent Platforms** (Agent orchestration, reasoning, tool use)
- **Business Process Automation** (Workflow orchestration, triggers, approvals)
- **Enterprise Knowledge Management** (RAG, document intelligence, grounding)
- **AI Observability & Governance** (Monitoring, evaluation, audit, guardrails)

1.3 Core Value Proposition

"Replace repetitive, costly human workflows with an AI-powered digital workforce that reasons, retrieves information, uses tools, handles complex decisions, and escalates to humans only when necessary — all within a governed, auditable, and observable platform."

1.4 Primary Differentiators

1. **End-to-End Workflow Automation** — From trigger to completion, including AI reasoning, knowledge retrieval, tool use, branching logic, and human approvals in a single platform.
 2. **Human-in-the-Loop by Design** — Not bolted on. The platform architecturally enforces that high-risk or ambiguous actions require human approval.
 3. **Multi-Tenant with Strong Isolation** — Enterprise-grade multi-tenancy with strict data and operational isolation between organizations.
 4. **AI Observability First** — Full execution traces, token usage, cost tracking, latency, AI evaluation, and hallucination monitoring built in from day one.
 5. **Governance & Guardrails** — Risk classification, policy enforcement, approval workflows, and audit trails as first-class platform citizens.
 6. **Vendor-Agnostic AI** — Support for multiple AI model providers, preventing vendor lock-in and enabling cost optimization.
-

2. PRODUCT VISION & MISSION

2.1 Product Vision

"To become the operating system for AI-powered business — where every organization, regardless of size or industry, can build, deploy, and govern an intelligent digital workforce that augments human capability at scale."

2.2 Product Mission

"To democratize enterprise-grade AI workflow automation by providing a platform where non-technical business users and developers alike can create AI agents, connect organizational knowledge, build automated workflows, maintain human oversight, and trust that every AI action is auditable, explainable, and governed."

2.3 Core Platform Principles

The platform is designed around the following immutable principles:

1. **Multi-Tenancy:** Every feature, data record, and operation is scoped to an organization. No data leaks across organization boundaries.
2. **Security-First:** Security is not an afterthought. Access control, encryption, and authorization are enforced at every layer.
3. **Privacy-First:** Organizational data, user data, and AI inputs/outputs are handled with strict privacy principles.
4. **Least-Privilege Access:** Users, AI agents, and integrations receive only the minimum permissions necessary to perform their function.
5. **Human-in-the-Loop:** Autonomy is a spectrum. The platform enforces human approval for high-risk, irreversible, or ambiguous AI actions.
6. **Explainability & Traceability:** Every AI decision, every workflow step, and every tool invocation must be traceable and explainable.
7. **AI Observability:** AI behavior is monitored, evaluated, and measured — not treated as a black box.
8. **AI Evaluation:** AI output quality is continuously measured against defined criteria.
9. **Workflow Reliability:** Workflows must be fault-tolerant, retry-capable, and idempotent where feasible.
10. **Fault Tolerance:** The platform must gracefully handle AI failures, tool failures, and external service unavailability.
11. **Retry & Recovery:** Failed operations must support intelligent retry strategies with configurable backoff.
12. **Idempotency:** Workflow steps must be designed to be safely re-executable where possible.
13. **Auditability:** Every significant action must produce an immutable audit event.
14. **Extensibility:** New agent types, tools, integrations, and workflow node types must be addable without breaking existing functionality.
15. **Modular Architecture:** Platform capabilities must be independently deployable and scalable modules.
16. **Cost Awareness:** Token usage, model costs, and execution costs must be tracked and surfaced to organizations.
17. **Production Scalability:** The platform must scale to handle hundreds of organizations and thousands of concurrent workflow executions.

18. **Vendor/Model Independence:** Organizations can switch AI models without redesigning workflows.
 19. **Enterprise Readiness:** SSO, RBAC, audit logs, compliance controls, SLA guarantees, and admin controls are built-in.
 20. **Developer-Friendly Extensibility:** API access, webhooks, and programmatic control allow developers to extend and integrate the platform.
-

3. PROBLEM STATEMENT

3.1 The Core Business Problem

Modern businesses face an ever-increasing volume of repetitive, knowledge-intensive operational work that requires reasoning, decision-making, document understanding, and system interaction. Traditional automation tools (RPA, workflow engines, rule-based systems) fail when tasks require:

- Natural language understanding
- Contextual reasoning
- Unstructured document comprehension
- Dynamic decision-making based on incomplete information
- Multi-step task planning and execution

AI technologies (LLMs, RAG, Agents) now provide the reasoning capability to automate these tasks. However, businesses are unable to operationalize AI because:

1. **No Unified Platform** — AI tools exist in silos. RAG pipelines, agent frameworks, workflow tools, and monitoring systems are disconnected.
2. **No Governance** — AI agents acting autonomously without guardrails, approval policies, or audit trails create business and compliance risk.
3. **No Observability** — Businesses cannot see what their AI is doing, why it made a decision, or whether its outputs are correct.
4. **No Trust** — Without HITL controls, organizations fear deploying AI in production for consequential tasks.
5. **No Enterprise-Grade Security** — Most AI tools lack proper multi-tenancy, RBAC, data isolation, and security controls.
6. **No Integration Depth** — AI cannot act on the world without secure, reliable connections to business systems (CRMs, ticketing, email, databases).
7. **High Technical Barrier** — Building a production AI agent system requires deep technical expertise not available in most organizations.

3.2 Current State Pain Points

Pain Point	Impact
Manual knowledge work is bottlenecking operations	High headcount cost, slow throughput
AI PoCs fail to reach production	Wasted investment, lack of governance and reliability
Business users cannot create or manage AI agents	High dependency on engineering teams
AI acts autonomously on high-risk tasks	Compliance risk, data loss, reputational damage
No visibility into AI reasoning	No explainability, no debugging, no improvement
Tool integration requires custom engineering	Slow time-to-value, high maintenance cost
No way to measure AI quality over time	Hallucinations and failures go undetected

Pain Point	Impact
Different tools for different use cases	Platform sprawl, training overhead, maintenance cost

4. BUSINESS OPPORTUNITY

4.1 Market Context

The global AI automation and workflow intelligence market is experiencing significant growth, driven by enterprise adoption of large language models, multimodal AI, and agentic systems. Businesses across industries are seeking platforms that allow them to:

- Convert AI prototypes into reliable production systems
- Automate knowledge-intensive business processes at scale
- Maintain human oversight over consequential AI actions
- Measure and improve AI output quality over time

4.2 Business Opportunity Definition

AI Workforce addresses a gap in the market for a **unified, enterprise-grade, multi-tenant AI workflow automation platform** that bridges the gap between:

- AI capability (LLMs, RAG, Agents)
- Business process automation (Workflow orchestration, approvals)
- Enterprise governance (Security, audit, compliance, HITL)

4.3 Revenue Model Considerations (Requirements Level)

The platform must be designed to support:

- A **subscription-based SaaS model** (monthly/annual plans)
- **Usage-based components** (workflow executions, token usage, storage)
- **Tiered plans** (Starter, Professional, Enterprise) with different feature sets and usage limits
- **Organization-level billing** with usage tracking per organization

Note: Specific pricing is a business decision, not an engineering requirement. The platform must be capable of tracking and enforcing usage limits and plan boundaries.

5. TARGET USERS & ORGANIZATIONS

5.1 Target Organizations

Primary Targets:

Organization Type	Description
Mid-Market Companies (50–500 employees)	Companies with operational complexity but limited AI engineering capacity. Seek to automate knowledge work without building custom AI infrastructure.
Product-Based SaaS Companies	Technology companies that want to embed AI automation into their own product operations or offer AI-powered features to their customers.
Service-Based Companies	Consulting firms, agencies, and professional services organizations with high-volume client-facing knowledge work (reports, onboarding, support).
Mid-Level Startups (Series A–C)	Startups scaling their operations and needing to automate processes that were previously handled manually at small team size.

Secondary Targets:

Organization Type	Description
Enterprise Divisions	Large enterprise internal divisions or innovation teams seeking AI automation without full IT dependency.
Digital-Native SMBs	Small but technically sophisticated businesses with automation-first culture.

5.2 Primary User Functions

- Business Operations Leaders
- Product Managers
- Customer Success Managers
- IT Operations Managers
- Knowledge Managers / Content Managers
- Sales Operations Managers
- HR Operations Managers
- AI/Automation Engineers
- Platform Administrators
- Executive Stakeholders (analytics consumers)

6. USER PERSONAS

Persona 1: Priya — Operations Manager

Attribute	Details
Role	Operations Manager
Technical Level	Non-technical
Organization Type	Service-based company, 150 employees
Primary Goal	Automate client onboarding and internal report generation
Pain Points	Repetitive manual tasks, slow document processing, no visibility into workflow status
Platform Usage	Creates workflows using the visual builder, monitors execution results, manages approval queues
Key Needs	Simple workflow creation, execution visibility, approval management

Persona 2: Arjun — AI/Automation Engineer

Attribute	Details
Role	AI Engineer / Platform Engineer
Technical Level	Highly technical
Organization Type	Product-based SaaS company, 80 employees
Primary Goal	Build reliable AI agents connected to company tools and knowledge base
Pain Points	Lack of observability, difficult to debug AI agent failures, no governance framework
Platform Usage	Creates and configures agents, connects integrations, monitors execution traces, evaluates AI output quality
Key Needs	Deep execution traces, evaluation framework, tool integration management, API access

Persona 3: Meera — Business Unit Admin

Attribute	Details
Role	Business Unit Administrator
Technical Level	Semi-technical
Organization Type	Mid-market company, 300 employees

Attribute	Details
Primary Goal	Manage the team's AI platform access, ensure compliance and appropriate usage
Pain Points	No visibility into who is using AI and for what, cannot enforce access policies
Platform Usage	Manages users, roles, permissions, reviews audit logs, configures organization-level settings
Key Needs	RBAC management, audit trails, usage visibility, policy enforcement

Persona 4: David — Customer Support Lead

Attribute	Details
Role	Customer Support Lead
Technical Level	Non-technical
Organization Type	Product-based SaaS company
Primary Goal	Use AI-powered agents to resolve customer support tickets faster
Pain Points	High ticket volume, slow response times, repetitive resolution paths
Platform Usage	Monitors AI agent support workflows, reviews AI-generated responses before sending, approves escalations
Key Needs	Human approval queue, AI draft review, clear escalation rules

Persona 5: Ravi — Executive / Decision Maker

Attribute	Details
Role	VP of Operations / CTO
Technical Level	Strategic level
Organization Type	Mid-market startup, Series B
Primary Goal	Understand ROI of AI automation, ensure safe and compliant deployment
Pain Points	No consolidated view of AI activity, cost overruns from AI model usage, governance concerns
Platform Usage	Reviews analytics dashboards, approves high-risk workflow policies, monitors AI spend
Key Needs	Executive dashboards, cost tracking, governance visibility

Persona 6: Nadia — Developer / API Consumer

Attribute	Details
Role	Backend Developer / Integration Engineer

Attribute	Details
Technical Level	Highly technical
Organization Type	Product-based company
Primary Goal	Trigger workflows programmatically and integrate AI Workforce into existing systems
Pain Points	Platform doesn't expose APIs, no webhook support, cannot trigger AI workflows from existing systems
Platform Usage	Uses API keys, triggers workflows via REST API, receives webhook notifications
Key Needs	Well-documented APIs, secure API key management, reliable webhook delivery

7. BUSINESS GOALS & PRODUCT GOALS

7.1 Business Goals

ID	Business Goal
BG-01	Establish a commercially viable SaaS platform that generates recurring subscription and usage revenue
BG-02	Achieve strong enterprise adoption by meeting security, compliance, and governance requirements
BG-03	Reduce the time-to-value for organizations deploying AI automation from months to days
BG-04	Enable organizations to quantifiably demonstrate ROI through automation metrics
BG-05	Build a platform extensible enough to support multiple industries and use cases without custom development
BG-06	Create a defensible competitive position through depth of governance, observability, and HITL capabilities
BG-07	Establish the platform as the “AI automation operating system” for mid-market and scaling companies

7.2 Product Goals

ID	Product Goal
PG-01	Enable any organization to create, configure, and deploy AI agents without AI engineering expertise
PG-02	Provide a visual workflow builder that non-technical users can use to build production-grade automation
PG-03	Enforce human oversight for high-risk AI actions through configurable approval workflows
PG-04	Deliver full AI observability: traces, token usage, cost, latency, and evaluation scores
PG-05	Enable organizations to connect their knowledge base and external tools to AI agents
PG-06	Provide a governance and audit framework that satisfies enterprise compliance requirements
PG-07	Support multiple AI model providers with the ability to switch without redesigning workflows
PG-08	Enable programmatic access and extensibility through a developer API

8. SUCCESS CRITERIA & KPIS

8.1 Platform Success Criteria

Category	Metric	Target
Adoption	Organizations onboarded (6 months post-launch)	≥ 50 organizations
Engagement	Workflows executed per week per active organization	≥ 100
Reliability	Workflow execution success rate	≥ 99.5%
Performance	Median workflow step execution latency	≤ 2 seconds (excluding AI inference)
Availability	Platform uptime SLA	≥ 99.9%
AI Quality	Agent evaluation pass rate	≥ 85% on defined evaluation suites
Security	Zero cross-tenant data leakage incidents	0 incidents
Time-to-Value	Time to deploy first workflow from sign-up	≤ 1 business day

8.2 Business KPIs

KPI	Description
Monthly Recurring Revenue (MRR)	Subscription revenue per month
Workflow Execution Volume	Total workflow runs across all organizations
Active Organizations	Organizations with at least 1 workflow execution in the period
AI Cost per Workflow	Average AI token cost per workflow execution
Human Approval Rate	Percentage of workflow executions requiring human approval
Automation Rate	Percentage of workflow executions completed without human intervention
Evaluation Score Trend	Average AI evaluation score over time (improving or degrading)
Time Saved	Estimated time saved per organization through automation
Net Promoter Score (NPS)	User satisfaction metric

9. SCOPE DEFINITION

9.1 In-Scope

The following capabilities are in scope for the AI Workforce platform across all planned phases:

- Multi-tenant organization and workspace management
- Identity, authentication, and access management
- Role-based access control (RBAC)
- AI agent creation, configuration, versioning, and lifecycle management
- Knowledge management: document upload, ingestion, indexing, and RAG retrieval
- Tool registry: integration with external APIs, databases, email, CRM, and ticketing systems
- Visual workflow builder: triggers, nodes, conditions, approvals, and actions
- Agentic orchestration: planning, tool selection, multi-step execution
- Human-in-the-loop (HITL): approval queues, escalation, delegation
- Guardrails and governance: risk classification, policy enforcement, data access policies
- Workflow execution engine: runs, status tracking, retries, failure handling
- AI observability: execution traces, logs, latency, token usage, cost
- AI evaluation: output quality, groundedness, hallucination monitoring
- Analytics: workflow metrics, agent metrics, cost metrics, usage trends
- Administration: user management, organization settings, usage limits, audit logs
- Developer API: programmatic workflow triggering, webhook notifications, API key management
- Notification system: in-app and email notifications for workflow events, approvals, and failures
- Multi-model AI support: configurable AI model provider and model selection per agent
- Audit trail: immutable record of all significant platform actions

9.2 Out of Scope (All Phases)

The following are explicitly out of scope for this platform:

- Building or hosting AI models (the platform consumes models from providers; it does not train or host them)
- Providing a general-purpose chat interface for end consumers (this is a workflow automation platform, not a consumer chatbot)
- Replacing core business systems (CRM, ERP, HRMS) — the platform integrates with them
- Real-time voice or video AI processing
- Mobile-native applications (mobile-responsive web is in scope; native iOS/Android apps are not)
- Providing AI model training, fine-tuning, or custom model hosting infrastructure
- Building a code generation or software development AI tool (unless it is a workflow use case)
- Consumer-facing or B2C product features

9.3 Phasing Summary

Capability	MVP	Phase 2	Phase 3
Auth & Identity	✓	✓	✓
Organization Management	✓	✓	✓
RBAC	✓	✓	✓
AI Agent (basic)	✓	✓	✓
Agent Versioning		✓	
Knowledge Upload & RAG	✓	✓	✓
Knowledge Versioning		✓	
Tool Registry (built-in tools)	✓	✓	
Custom OAuth Integrations		✓	
Visual Workflow Builder	✓	✓	✓
Human-in-the-Loop	✓	✓	
Guardrails & Governance	Basic	Advanced	Enterprise
Workflow Execution Engine	✓	✓	✓
Observability & Tracing	✓	✓	✓
AI Evaluation	Basic	Full	Enterprise
Analytics	Basic	Advanced	Enterprise
Developer API		✓	
Webhooks		✓	
SSO / SAML			✓
Multi-model Management	Basic	Advanced	
Audit Logs	✓	✓	✓
Billing/Plan Enforcement	Basic	Full	Enterprise

10. ASSUMPTIONS, CONSTRAINTS & DEPENDENCIES

10.1 Assumptions

ID	Assumption
A-01	AI model providers (e.g., OpenAI, Anthropic, Google, etc.) provide reliable, documented, and commercially available APIs
A-02	Organizations will bring their own AI model API credentials for some configurations
A-03	Organization administrators are technically capable of managing RBAC and platform settings
A-04	External tools and systems provide integration APIs (REST, webhooks, OAuth) for connectivity
A-05	Organizations will provide their own documents and knowledge assets for ingestion
A-06	The platform will operate in the cloud and be accessed via web browser
A-07	Data residency requirements will be addressed in Phase 3 (not MVP)
A-08	The platform's initial target geography is English-language markets; i18n is a future concern
A-09	Organizations agree to acceptable use policies governing AI usage on the platform

10.2 Constraints

ID	Constraint
C-01	The platform must not store AI model credentials in plaintext under any circumstances
C-02	Cross-tenant data access is architecturally prohibited — no exceptions
C-03	The platform must support standard web browsers (Chrome, Firefox, Edge, Safari) without plugins
C-04	All AI model interactions must be logged for audit purposes, subject to configured retention policies
C-05	The platform must be deployable on major cloud providers (architecture-agnostic requirement)
C-06	The MVP must be achievable by a small engineering team within a reasonable development cycle
C-07	Third-party services used must have commercially acceptable SLAs and data processing agreements

10.3 Dependencies

ID	Dependency	Type
D-01	AI model provider APIs (OpenAI, Anthropic, Google Vertex AI, etc.)	External — Critical
D-02	Vector database for knowledge storage and retrieval	Technical — Critical
D-03	Document processing pipelines for file ingestion	Technical — Critical
D-04	Email delivery service for notifications and invitations	External — High
D-05	OAuth provider support for external tool integrations	External — Medium
D-06	Storage service for uploaded documents	Technical — Critical
D-07	Authentication infrastructure	Technical — Critical

11. USER ROLES & PERMISSION CONCEPTS

11.1 Platform-Level Roles

Role	Description
Super Admin	Platform-level role. Manages all organizations, system settings, and platform health. Only internal team members.

11.2 Organization-Level Roles

Role	Description
Organization Owner	Full control over organization. Can delete organization, manage billing, assign all roles. One per organization (transferable).
Organization Admin	Administrative control. Can manage members, roles, settings, integrations, and all platform features within the organization.
Workflow Manager	Can create, edit, publish, and manage workflows and agents within assigned workspaces. Cannot manage users or billing.
Agent Builder	Can create and configure AI agents and knowledge bases. Cannot publish workflows or manage org settings.
Analyst	Read-only access to analytics, monitoring, execution history, and audit logs. Cannot create or modify anything.
Approver	Can view and act on approval queues. Cannot create agents or workflows.
Member	Basic user. Can view assigned workflows and trigger manual workflow executions. Cannot create or configure.
API User	Machine-to-machine role for programmatic access. Scoped to specific capabilities.

11.3 Permission Concepts

The platform uses **Role-Based Access Control (RBAC)** with the following principles:

- 1. Organization Scoping:** All permissions are scoped to an organization. A user may have different roles in different organizations.
- 2. Workspace Scoping (Phase 2):** Permissions may be further scoped to specific workspaces within an organization.
- 3. Resource-Level Control:** Certain resources (agents, workflows, knowledge collections) may have additional access controls defining which roles can read, write, execute, or delete them.
- 4. Deny by Default:** If a permission is not explicitly granted, it is denied.
- 5. Role Inheritance:** Higher-level roles inherit permissions of lower-level roles within the same scope.
- 6. Agent Permissions:** AI agents do not have human user roles. Agents have capability configurations (which tools they can use, which knowledge they can access) defined by authorized administrators.

11.4 Permission Domains

Domain	Description
Identity	Sign in, sign out, manage own profile, manage MFA
Organization	View org, edit org settings, manage billing, delete org
Members	Invite users, remove users, assign roles
Agents	Create, read, update, delete, publish, test agents
Knowledge	Create, upload, read, update, delete, re-index knowledge
Workflows	Create, read, update, delete, publish, execute, disable workflows
Tools	Register, configure, read, delete tools and credentials
Approvals	View approval queue, approve, reject, delegate, escalate
Monitoring	View execution logs, traces, metrics
Evaluation	Create, run, view evaluation suites and results
Analytics	View analytics dashboards and reports
Audit Logs	View audit events
API Keys	Create, view, revoke API keys
Administration	Manage organization-level policies, guardrail settings

12. MULTI-TENANT ARCHITECTURE REQUIREMENTS

12.1 Tenancy Model

- The platform is designed for a **true multi-tenant SaaS model** where all organizations share a common platform infrastructure.
- Each organization is a logically isolated tenant.
- All data — users, agents, workflows, knowledge, executions, logs — is strictly scoped to one organization.
- No organization can access, query, or in any way observe another organization's data.
- Isolation must be enforced at the data layer, application layer, and API layer.

12.2 Multi-Tenant Requirements

ID	Requirement
MT-01	Every data entity must carry an organization identifier that is enforced in all read/write operations
MT-02	API endpoints must validate that the requesting user belongs to the target organization
MT-03	An organization's knowledge collections must not be accessible to agents of another organization
MT-04	AI execution logs and traces must be isolated per organization
MT-05	Organization-level usage limits must be enforced independently per organization
MT-06	Storage buckets, vector namespaces, and databases must be logically isolated per organization
MT-07	Organization settings, guardrails, and policies are independent per organization
MT-08	System administrators must be able to view cross-organization platform health without accessing tenant data
MT-09	A user may belong to multiple organizations with different roles in each
MT-10	An organization must be able to be suspended or deleted without affecting other organizations

12.3 Workspaces (Phase 2)

Within a single organization, **Workspaces** provide a second level of logical grouping that allows organizations to create separate environments (e.g., by department, team, or project). Workspaces inherit organization-level policies but may have additional restrictions.

13. FUNCTIONAL REQUIREMENTS

13.1 Identity & Access Management (IAM)

13.1.1 REGISTRATION & AUTHENTICATION

ID	Requirement
IAM-01	The platform must support user self-registration with email and password
IAM-02	Registered users must verify their email address before accessing platform features
IAM-03	Email verification links must expire after a configurable period (default 24 hours)
IAM-04	The platform must support secure password-based authentication
IAM-05	Passwords must meet configurable strength requirements (minimum length, complexity)
IAM-06	The platform must support Multi-Factor Authentication (MFA) via TOTP (Phase 1)
IAM-07	The platform must support SSO via SAML 2.0 (Phase 3) and OAuth 2.0 (Phase 3)
IAM-08	Sessions must be managed with secure, configurable expiry and idle timeout
IAM-09	The platform must support "remember this device" functionality for trusted devices
IAM-10	Users must be able to log out from all sessions simultaneously

13.1.2 PASSWORD MANAGEMENT

ID	Requirement
IAM-11	Users must be able to request a password reset via email
IAM-12	Password reset links must be single-use and expire after a configurable period
IAM-13	Password changes must invalidate all existing sessions
IAM-14	The system must enforce a password history policy preventing reuse of recent passwords
IAM-15	Account lockout must occur after a configurable number of failed login attempts

13.1.3 ACCOUNT LIFECYCLE

ID	Requirement
IAM-16	Users can be deactivated by an organization admin, preventing login while preserving data
IAM-17	Deactivated users must lose access to all organization resources immediately
IAM-18	Users can be permanently deleted, subject to data retention and audit requirements

ID	Requirement
IAM-19	An organization owner can transfer ownership to another member
IAM-20	A user who is the sole owner of an organization must transfer or delete the organization before deleting their account

13.2 Organization & Workspace Management

13.2.1 ORGANIZATION MANAGEMENT

ID	Requirement
ORG-01	Authenticated users can create a new organization by providing a name and profile
ORG-02	The creating user automatically becomes the organization owner
ORG-03	Organizations must have a unique, human-readable slug/identifier
ORG-04	An organization owner or admin can update organization name, description, logo, and settings
ORG-05	An organization owner can delete the organization, which triggers a cascading soft-delete or hard-delete per data retention policy
ORG-06	Organization deletion must require explicit confirmation to prevent accidental loss
ORG-07	An organization can be suspended by a Super Admin (e.g., for non-payment), preventing login for all members

13.2.2 MEMBER INVITATIONS

ID	Requirement
ORG-08	Organization admins can invite users by email address
ORG-09	Invited users receive an email invitation with a time-limited, single-use acceptance link
ORG-10	If the invited user does not have an account, they must register before accepting the invitation
ORG-11	Admins can revoke pending invitations
ORG-12	Admins can resend expired invitations
ORG-13	Admins can set the role to be assigned to the invited user at invitation time
ORG-14	A user can belong to multiple organizations simultaneously

13.2.3 MEMBER MANAGEMENT

ID	Requirement
ORG-15	Admins can view all members of the organization and their roles
ORG-16	Admins can change the role of any member (except the owner)
ORG-17	Admins can remove members from the organization (member loses access; their data contributions remain)
ORG-18	An organization owner cannot be removed; ownership must be transferred first

13.2.4 ORGANIZATION SETTINGS

ID	Requirement
ORG-19	Organizations must be able to configure AI model providers and default models
ORG-20	Organizations must be able to set usage limits (workflow runs, token usage, storage)
ORG-21	Organizations must be able to configure default approval policies
ORG-22	Organizations must be able to configure session and security policies (MFA enforcement, session timeout)
ORG-23	Organizations must be able to configure data retention policies within platform limits

13.3 AI Agent Management

13.3.1 AGENT CREATION & CONFIGURATION

ID	Requirement
AGT-01	Authorized users can create a new AI agent within their organization
AGT-02	An agent must have a name, description, purpose statement, and system instructions
AGT-03	An agent must be configured with a specific AI model provider and model
AGT-04	An agent can be configured with model-level parameters (e.g., temperature, max tokens)
AGT-05	An agent must have a defined set of tool permissions — specifying which tools it is allowed to invoke
AGT-06	An agent must have a defined set of knowledge access permissions — specifying which knowledge collections it can query

ID	Requirement
AGT-07	An agent must have a defined memory strategy (stateless, session-scoped, long-term)
AGT-08	An agent must have a defined maximum execution budget (max steps, max tokens, max duration)
AGT-09	An agent can be configured with output format expectations (structured JSON, free text, etc.)
AGT-10	An agent must have a defined risk profile (Low, Medium, High, Critical)

13.3.2 AGENT LIFECYCLE

ID	Requirement
AGT-11	Agents exist in the following states: Draft, Testing, Published, Deprecated, Archived
AGT-12	An agent in Draft state can be edited freely
AGT-13	An agent must pass a configuration validation check before it can move to Testing state
AGT-14	An agent in Testing state can be executed in a sandboxed test environment
AGT-15	An agent can be Published only by users with appropriate permissions
AGT-16	Only Published agents can be used in active workflow executions
AGT-17	An agent can be Deprecated to prevent new workflows from using it while existing workflows continue
AGT-18	An agent can be Archived, which removes it from active lists but preserves historical data

13.3.3 AGENT VERSIONING

ID	Requirement
AGT-19	Every change to an agent configuration creates a new version (Phase 2)
AGT-20	Versions must be immutable once created
AGT-21	Users can view the history of agent versions
AGT-22	Users can roll back an agent to a previous version
AGT-23	Active workflows using a specific agent version continue to use that version until explicitly updated
AGT-24	A version must carry a changelog entry describing what changed

13.3.4 AGENT TESTING

ID	Requirement
AGT-25	An agent can be tested in isolation using a test input, producing a test execution result
AGT-26	Test executions are isolated from production and do not affect production data or tools
AGT-27	Test execution results must display the agent's reasoning trace, tool calls, knowledge retrievals, and output
AGT-28	Administrators can configure test-specific tool mocks or use real tools in test mode

13.4 Knowledge Management & RAG

13.4.1 KNOWLEDGE SOURCES & COLLECTIONS

ID	Requirement
KNW-01	Organizations can create named Knowledge Collections to organize related documents
KNW-02	Knowledge Collections must have defined access permissions (which agents and users can access them)
KNW-03	A Knowledge Collection must have a name, description, and owner
KNW-04	Organizations can create multiple Knowledge Collections for different domains or purposes

13.4.2 DOCUMENT INGESTION

ID	Requirement
KNW-05	Users can upload documents to a Knowledge Collection (PDF, DOCX, TXT, MD, HTML supported)
KNW-06	Uploaded documents must go through an asynchronous ingestion pipeline
KNW-07	Ingestion pipeline must: extract text, chunk content, generate embeddings, and index in the vector store
KNW-08	Users must be able to see the ingestion status of each document (Pending, Processing, Indexed, Failed)
KNW-09	If ingestion fails, the user must receive a clear error and be able to retry
KNW-10	Users can add metadata to documents (tags, category, source URL, effective date)
KNW-11	Users can add knowledge from URLs (web page content ingestion)
KNW-12	Users can add knowledge from plain text input

13.4.3 DOCUMENT LIFECYCLE

ID	Requirement
KNW-13	Users can update a document by re-uploading a new version; the old version is archived or replaced per policy

ID	Requirement
KNW-14	Users can delete a document, which must trigger de-indexing from the vector store
KNW-15	Deleted documents must not be retrievable in RAG queries
KNW-16	Users can trigger a re-indexing of a Knowledge Collection to rebuild embeddings
KNW-17	Documents must carry source traceability metadata so retrieved content can be traced to its source

13.4.4 RAG RETRIEVAL

ID	Requirement
KNW-18	AI agents must be able to retrieve relevant context from permitted Knowledge Collections during execution
KNW-19	Retrieval must be based on semantic similarity to the agent's query
KNW-20	Retrieval results must include source attribution (document name, page, section)
KNW-21	Retrieval results must be configurable (top-K results, similarity threshold)
KNW-22	RAG queries must be scoped to the Knowledge Collections the agent is permitted to access
KNW-23	Retrieval results must be logged as part of the agent's execution trace
KNW-24	The system must support hybrid retrieval strategies (semantic + keyword) in Phase 2

13.4.5 KNOWLEDGE ACCESS CONTROL

ID	Requirement
KNW-25	A Knowledge Collection can be configured as organization-wide or restricted to specific agents/roles
KNW-26	An agent can only access Knowledge Collections it has been explicitly granted access to
KNW-27	Users can only manage Knowledge Collections they have been granted permission to manage

13.5 Tools & Integrations

13.5.1 TOOL REGISTRY

ID	Requirement
TOOL-01	The platform maintains a Tool Registry of available tools that agents can invoke

ID	Requirement
TOOL-02	The platform provides a set of built-in tools (email, HTTP request, etc.)
TOOL-03	Organizations can register custom tools via REST API specifications
TOOL-04	Each tool in the registry must have: name, description, input schema, output schema, execution policy
TOOL-05	Tools must be categorized (Email, CRM, Database, Ticketing, HTTP, Webhook, etc.)
TOOL-06	Tools must have a defined risk level (Low, Medium, High, Critical) affecting approval requirements
TOOL-07	The tool registry must be searchable and filterable by category and risk level

13.5.2 BUILT-IN TOOLS (MVP)

ID	Requirement
TOOL-08	The platform must provide a built-in HTTP/REST API tool for calling external endpoints
TOOL-09	The platform must provide a built-in email sending tool
TOOL-10	The platform must provide a built-in web search tool (Phase 2)
TOOL-11	The platform must provide a built-in data transformation/extraction tool

13.5.3 INTEGRATION CONNECTORS (PHASE 2)

ID	Requirement
TOOL-12	The platform must support OAuth 2.0 integration with external services
TOOL-13	Pre-built connectors must be available for common business systems (CRM, ticketing, email platforms)
TOOL-14	Each connector must handle OAuth token refresh automatically
TOOL-15	Connector credentials must be stored encrypted and scoped to the organization

13.5.4 API CREDENTIAL MANAGEMENT

ID	Requirement
TOOL-16	Organizations can store API credentials (API keys, tokens, secrets) in a secure credential vault
TOOL-17	Stored credentials must never be returned in plaintext via the UI or API
TOOL-18	Credentials must be encrypted at rest using organization-specific encryption keys
TOOL-19	Credentials must be scoped to specific tools or all tools within an organization
TOOL-20	Credential access must be logged as an audit event

13.5.5 TOOL EXECUTION POLICY

ID	Requirement
TOOL-21	Tool invocations by agents must be logged with input, output, latency, and status
TOOL-22	Tools must have configurable retry policies (max retries, backoff strategy)
TOOL-23	Tool failures must be classified as transient (retryable) or permanent (fail workflow step)
TOOL-24	High-risk tool invocations must require human approval before execution
TOOL-25	Tool execution must be subject to configurable rate limits per organization
TOOL-26	Tool responses must be validated against the tool's defined output schema

13.6 Workflow Automation

13.6.1 WORKFLOW CREATION & STRUCTURE

ID	Requirement
WF-01	Authorized users can create a new workflow within their organization
WF-02	A workflow must have a name, description, purpose, and owner
WF-03	A workflow is composed of a directed graph of nodes connected by edges
WF-04	Nodes represent actions, decisions, or control points in the workflow
WF-05	Edges represent the flow of control from one node to another
WF-06	Every workflow must have exactly one Start node (trigger)
WF-07	A workflow can have one or more End nodes
WF-08	The workflow structure must be validated for completeness before publishing

13.6.2 NODE TYPES

ID	Node Type	Description
WF-09	Trigger Node	The entry point of a workflow; defines how the workflow is initiated
WF-10	AI Agent Node	Invokes a configured AI agent with defined input; receives agent output
WF-11	RAG Retrieval Node	Queries a knowledge collection and returns retrieved context
WF-12	Tool Node	Invokes a registered tool with defined parameters
WF-13	Condition Node	Evaluates a logical condition on workflow data; routes flow to different branches
WF-14	Human Approval Node	Pauses the workflow and requests human approval before continuing

ID	Node Type	Description
WF-15	Transform Node	Transforms or maps data between workflow steps
WF-16	Wait/Delay Node	Pauses workflow execution for a defined duration or until a condition is met
WF-17	Notification Node	Sends a notification (email, in-app) as part of the workflow
WF-18	Sub-Workflow Node	Invokes another published workflow as a sub-process (Phase 2)
WF-19	Loop Node	Iterates over a list of items, executing contained nodes for each item (Phase 2)
WF-20	End Node	Marks the completion of a workflow path

13.6.3 TRIGGER TYPES

ID	Trigger Type	Description
WF-21	Manual Trigger	Workflow is started by a user explicitly from the UI
WF-22	API Trigger	Workflow is started via a REST API call (Phase 2)
WF-23	Scheduled Trigger	Workflow runs on a defined schedule (cron expression or natural language)
WF-24	Webhook Trigger	Workflow is started by an incoming webhook from an external system
WF-25	Event Trigger	Workflow is started by a platform-internal event (e.g., a new document uploaded) (Phase 2)

13.6.4 WORKFLOW LIFECYCLE

ID	Requirement
WF-26	Workflows exist in the following states: Draft, Published, Disabled, Archived
WF-27	A workflow in Draft state can be freely edited
WF-28	A workflow must pass structural validation before publishing
WF-29	Only Published workflows can be triggered for execution
WF-30	A Published workflow can be Disabled, stopping new executions while not deleting the workflow
WF-31	A Disabled workflow can be re-enabled (Published)
WF-32	An Archived workflow is hidden from active lists but preserved for historical reference

ID	Requirement
WF-33	Workflow versions must be created when a published workflow is modified (Phase 2)
WF-34	Running executions must complete on the version they started; version changes do not affect in-flight executions

13.6.5 VISUAL WORKFLOW BUILDER

ID	Requirement
WF-35	The platform must provide a visual, drag-and-drop workflow builder
WF-36	The workflow builder must support adding, removing, and repositioning nodes on a canvas
WF-37	The workflow builder must support drawing connections (edges) between nodes
WF-38	The workflow builder must display node configuration panels for editing node properties
WF-39	The workflow builder must support undo/redo operations
WF-40	The workflow builder must validate the workflow structure and highlight errors in real time
WF-41	The workflow builder must support copy/paste of nodes
WF-42	The workflow builder must display a read-only view for users without edit permissions

13.7 Agentic Orchestration

13.7.1 ORCHESTRATION REQUIREMENTS

ID	Requirement
ORCH-01	An AI agent node must receive its context (inputs, prior step outputs, retrieved knowledge) from the workflow execution state
ORCH-02	The orchestration engine must support multi-step agent execution: the agent can plan, execute tools, retrieve knowledge, and reason across multiple steps to complete a task
ORCH-03	The orchestration engine must track the agent's current state (step, tool calls, intermediate results) throughout execution
ORCH-04	The orchestration engine must enforce the agent's maximum execution budget (max steps, max tokens, max duration)
ORCH-05	If the agent exceeds its budget, the execution must fail gracefully with an explanation
ORCH-06	The orchestration engine must support agent-to-agent handoffs where one agent delegates to another (Phase 2)
ORCH-07	The orchestration engine must detect and handle infinite loop conditions

ID	Requirement
ORCH-08	The agent's reasoning trace (plan, tool calls, observations, final answer) must be captured in full

13.7.2 SAFE AUTONOMY

ID	Requirement
ORCH-09	The orchestration engine must classify each intended action (tool call) against the risk policy before execution
ORCH-10	High-risk or Critical actions must be routed to a Human Approval Node before execution
ORCH-11	The orchestration engine must not proceed with a High-risk action without an approval record
ORCH-12	The orchestration engine must support a "dry-run" mode that plans but does not execute actions (Phase 2)
ORCH-13	The orchestration engine must enforce tool permission checks: an agent cannot call a tool it is not permitted to use

13.7.3 CONTEXT MANAGEMENT

ID	Requirement
ORCH-14	The orchestration engine must manage the agent's context window, preventing token limit overflows
ORCH-15	For long-running or multi-step tasks, the engine must implement context summarization or windowing strategies
ORCH-16	The execution context must include: workflow inputs, prior step outputs, retrieved knowledge, tool results, and agent instructions

13.8 Human-in-the-Loop (HITL)

13.8.1 APPROVAL REQUESTS

ID	Requirement
HITL-01	When a workflow reaches a Human Approval Node, execution must pause and an approval request must be created
HITL-02	The approval request must include: the context that led to the approval request, the proposed action to be approved, and the submitting agent/workflow details
HITL-03	Approval requests must be routed to configured approvers or approval groups

ID	Requirement
HITL-04	Approvers must receive notifications (in-app and email) when an approval request is assigned to them
HITL-05	An approver can: Approve, Reject, or Request Changes on an approval request
HITL-06	An approver can optionally provide a comment or modification with their decision
HITL-07	When approved, the workflow must resume execution from the point it was paused
HITL-08	When rejected, the workflow must fail the current path and enter a defined failure handling state
HITL-09	When changes are requested, the workflow must route to a defined "revision" path if configured, or pause for re-submission

13.8.2 APPROVAL POLICIES

ID	Requirement
HITL-10	Organizations can define approval policies mapping risk levels to approval requirements
HITL-11	A policy can require a single approver or multiple approvers (quorum/unanimous)
HITL-12	A policy can define specific roles or users who are authorized to approve
HITL-13	A policy must define a timeout duration after which the approval escalates or auto-rejects

13.8.3 ESCALATION & DELEGATION

ID	Requirement
HITL-14	If an approval times out without action, it must automatically escalate to the next configured level
HITL-15	Approvers can delegate an approval to another authorized user
HITL-16	Delegation must be logged as an audit event
HITL-17	If no approver acts after all escalation levels are exhausted, the approval auto-rejects and the workflow fails safely

13.8.4 APPROVAL HISTORY

ID	Requirement
HITL-18	A complete history of all approval decisions must be maintained and auditable

ID	Requirement
HITL-19	Approval history must include: request details, approver identity, decision, comments, and timestamps
HITL-20	Approval history must be viewable by admins and analysts

13.9 Guardrails & Governance

13.9.1 RISK CLASSIFICATION

ID	Requirement
GRD-01	Every tool must be classified with a risk level: Low, Medium, High, Critical
GRD-02	Every agent must be assigned an overall risk profile based on its configured tools and purpose
GRD-03	Every workflow must have an effective risk level based on its highest-risk node
GRD-04	Risk levels must map to execution policies (autonomous, approval-required, blocked)
GRD-05	Organizations can customize the mapping of risk levels to execution policies

13.9.2 AI BEHAVIOR POLICIES

ID	Requirement
GRD-06	Organizations can define prohibited topics or action categories for their AI agents
GRD-07	Agents must be instructed not to process or expose PII beyond what is necessary for the task
GRD-08	Agents must have configurable guardrails to refuse requests that violate defined policies
GRD-09	The platform must support input validation (prompt injection detection awareness)
GRD-10	The platform must support output filtering for sensitive data patterns (PII, secrets)

13.9.3 DATA ACCESS POLICIES

ID	Requirement
GRD-11	An agent's access to knowledge and tools must be explicitly granted; no implicit access
GRD-12	Cross-knowledge collection access by a single agent must be explicitly permitted
GRD-13	Organizations can define data classification labels for knowledge collections
GRD-14	Agents configured with access to a restricted knowledge collection must be subject to stricter logging

13.9.4 AUDIT REQUIREMENTS

ID	Requirement
GRD-15	All significant platform actions must produce an immutable audit event
GRD-16	Audit events must include: timestamp, actor (user or system), action type, resource affected, organization, and outcome
GRD-17	Audit logs must be tamper-evident
GRD-18	Audit logs must be retained for a minimum configurable period (default 1 year)
GRD-19	Audit logs must be searchable and filterable by time, actor, action type, and resource
GRD-20	Audit logs must be exportable for compliance purposes

13.10 Execution & Monitoring

13.10.1 WORKFLOW EXECUTION

ID	Requirement
EXEC-01	A workflow execution (Run) must have a unique identifier and track its full lifecycle
EXEC-02	A Run must record: workflow ID, workflow version, trigger type, start time, end time, status, and triggering user/system
EXEC-03	Run statuses must include: Queued, Running, Paused (awaiting approval), Succeeded, Failed, Cancelled, Timed Out
EXEC-04	Each node within a Run must have a Step Execution record with: node ID, status, start time, end time, input, output, and error (if any)
EXEC-05	A Run can be cancelled by authorized users while it is in Queued, Running, or Paused state
EXEC-06	Cancellation must be graceful: in-progress tool calls or agent steps must be allowed to complete or safely terminated
EXEC-07	A Failed Run can be retried by authorized users, subject to idempotency considerations
EXEC-08	Retries must create a new Run record linked to the original failed Run
EXEC-09	The system must enforce a maximum concurrent run limit per organization based on their plan

ID	Requirement
EXEC-10	The system must enforce a maximum total run limit per period per organization based on their plan

13.10.2 EXECUTION LOGGING

ID	Requirement
EXEC-11	Every step execution must produce structured logs capturing all inputs, outputs, and intermediate states
EXEC-12	AI agent reasoning traces must be stored as part of the step execution log
EXEC-13	Tool invocation details must be logged (tool name, input, output, latency, status)
EXEC-14	Knowledge retrieval results must be logged (query, retrieved chunks, scores)
EXEC-15	Log retention must be configurable and subject to data retention policies
EXEC-16	Logs containing sensitive data must be masked based on organization-configured masking rules

13.10.3 COST & TOKEN TRACKING

ID	Requirement
EXEC-17	Token usage (prompt tokens, completion tokens) must be tracked per agent invocation
EXEC-18	Token usage must be aggregated per Step, per Run, per Workflow, per Agent, and per Organization
EXEC-19	Estimated AI cost must be calculated and stored based on model pricing
EXEC-20	Organizations must be able to view their token usage and AI cost in real time
EXEC-21	Organizations can configure cost budget alerts (threshold notifications)

13.10.4 FAILURE HANDLING

ID	Requirement
EXEC-22	Step-level failures must be classifiable: transient (retryable) or permanent (non-retryable)
EXEC-23	Workflow steps must support configurable retry policies (max attempts, backoff interval)
EXEC-24	After exhausting retries, a step failure must propagate to the Run level and mark the Run as Failed
EXEC-25	Workflows must support configurable failure handling paths (notify, compensate, fail fast)
EXEC-26	Organization admins and relevant roles must be notified of workflow failures

13.11 AI Evaluation

13.11.1 EVALUATION FRAMEWORK

ID	Requirement
EVAL-01	The platform must provide an AI Evaluation module for assessing agent output quality
EVAL-02	Organizations can create Evaluation Datasets: sets of input/expected output pairs
EVAL-03	Organizations can run an evaluation suite against a configured agent using an evaluation dataset
EVAL-04	Evaluation runs must be isolated from production executions
EVAL-05	Evaluation results must be stored and versioned for trend analysis

13.11.2 EVALUATION CRITERIA

ID	Requirement
EVAL-06	The platform must support evaluation of: Task Completion, Groundedness (answer supported by retrieved knowledge), Relevance, and Fluency
EVAL-07	The platform must support hallucination detection scoring (Phase 2)
EVAL-08	Organizations can define custom evaluation criteria (Phase 2)
EVAL-09	Evaluation scores must be viewable per agent, per model, and over time
EVAL-10	Organizations can configure evaluation score thresholds that trigger alerts when quality degrades

13.11.3 RAG EVALUATION

ID	Requirement
EVAL-11	The platform must support evaluation of RAG retrieval quality (recall, precision, relevance)
EVAL-12	Knowledge collection evaluation must identify poorly performing document segments

13.12 Observability

ID	Requirement
OBS-01	The platform must provide a centralized Observability dashboard per organization
OBS-02	The dashboard must display: active runs, recent failures, latency trends, token usage, and cost
OBS-03	AI execution traces must be viewable in a hierarchical format (Workflow → Step → Agent → Tool → RAG)

ID	Requirement
OBS-04	The platform must support error rate monitoring and alerting per workflow
OBS-05	The platform must expose execution latency metrics at the step and workflow level
OBS-06	The platform must support configurable alerts (threshold-based) for key metrics
OBS-07	Alerts must be deliverable via in-app notifications and email
OBS-08	Super Admins must have access to platform-level health metrics without accessing tenant data

13.13 Analytics & Reporting

ID	Requirement
ANA-01	The platform must provide an Analytics dashboard per organization
ANA-02	Analytics must include: total workflow executions, success rate, failure rate, avg. latency, and token usage
ANA-03	Analytics must be filterable by workflow, agent, date range, and triggering user
ANA-04	The platform must display human approval rate and automation rate metrics
ANA-05	The platform must display AI cost trends over time
ANA-06	The platform must display workflow execution volume trends
ANA-07	Organizations must be able to export analytics data in CSV format (Phase 2)
ANA-08	The platform must calculate and display estimated time saved through automation

13.14 Administration

ID	Requirement
ADM-01	Organization Admins have access to a dedicated Administration section
ADM-02	The Administration section must cover: Users, Roles, Integrations, AI Models, Policies, Usage, and Audit Logs
ADM-03	Admins can configure organization-level AI model settings (default model, cost limits)
ADM-04	Admins can configure organization-level guardrail policies
ADM-05	Admins can view and export audit logs
ADM-06	Admins can manage integration credentials

ID	Requirement
ADM-07	Admins can view usage metrics and remaining usage budget
ADM-08	Super Admins have a separate platform administration console for system-level management

13.15 Developer / API Platform (Phase 2)

ID	Requirement
DEV-01	The platform must expose a documented REST API for programmatic access
DEV-02	Authenticated users with appropriate permissions can create API keys for their organization
DEV-03	API keys must have configurable scopes (read-only, execute workflows, manage agents)
DEV-04	API keys must be revocable at any time
DEV-05	API key usage must be logged as audit events
DEV-06	The platform must support outbound webhooks notifying external systems of workflow events
DEV-07	Webhook endpoints must be configurable per organization
DEV-08	Webhook deliveries must be retried on failure with configurable backoff
DEV-09	Webhook delivery history must be viewable and debuggable

14. NON-FUNCTIONAL REQUIREMENTS

14.1 Security

ID	Requirement
SEC-01	All data in transit must be encrypted using TLS 1.2 or higher
SEC-02	All data at rest must be encrypted using AES-256 or equivalent
SEC-03	Authentication tokens must be short-lived (access token $\leq$ 15 minutes; refresh token $\leq$ 7 days)
SEC-04	API keys must be stored as hashed values; the full key is shown only once at creation
SEC-05	The platform must implement CSRF protection on all state-changing operations
SEC-06	The platform must implement rate limiting on authentication endpoints to prevent brute-force attacks
SEC-07	Input validation must be enforced on all API endpoints
SEC-08	Dependencies must be kept up to date to avoid known CVEs
SEC-09	The platform must support IP allowlisting for API access (Phase 2)

14.2 Privacy

ID	Requirement
PRIV-01	The platform must comply with GDPR principles for EU-based organizations
PRIV-02	Users must be able to request export of their personal data
PRIV-03	Users must be able to request deletion of their personal data
PRIV-04	AI inputs and outputs containing PII must be masked in logs per organization policy
PRIV-05	The platform must maintain a data processing record (Phase 3)

14.3 Reliability

ID	Requirement
REL-01	Platform availability SLA must be $\geq$ 99.9% (excluding planned maintenance)
REL-02	Workflow executions must be resumable after unexpected platform restarts
REL-03	The workflow execution engine must be fault-tolerant with no single point of failure
REL-04	Platform maintenance must be performable without complete downtime

ID	Requirement
REL-05	The platform must support graceful degradation: if a non-critical service is unavailable, core features should continue

14.4 Performance

ID	Requirement
PERF-01	API response time (P95) must be ≤ 500 ms for non-AI operations
PERF-02	The visual workflow builder must load and respond within 2 seconds for workflows up to 50 nodes
PERF-03	Knowledge retrieval (RAG query) must complete within 2 seconds for standard collection sizes
PERF-04	The platform must support at least 1,000 concurrent workflow executions per deployment
PERF-05	The platform must support at least 100 organizations with active workloads

14.5 Scalability

ID	Requirement
SCAL-01	The platform must scale horizontally to handle increasing load
SCAL-02	The workflow execution engine must be independently scalable from the API layer
SCAL-03	The knowledge ingestion pipeline must be independently scalable
SCAL-04	Database design must support efficient querying as data volume grows per organization

14.6 Availability

ID	Requirement
AVAIL-01	Planned maintenance windows must be communicated at least 24 hours in advance
AVAIL-02	The platform must implement automated health checks and self-healing capabilities
AVAIL-03	The platform must implement automatic failover for critical services

15. BUSINESS RULES

ID	Business Rule
BR-01	An agent CANNOT execute a High or Critical risk tool without an approved Human Approval Record
BR-02	A workflow CANNOT be published if it contains structural validation errors
BR-03	A user CANNOT access another organization's resources, regardless of their role within their own organization
BR-04	An organization's usage CANNOT exceed their plan limits; additional executions must be blocked or require plan upgrade
BR-05	An agent's tool invocations MUST be restricted to tools the agent is explicitly granted permission to use
BR-06	Knowledge retrieval MUST only return content from Knowledge Collections the agent is explicitly permitted to access
BR-07	Audit logs CANNOT be modified or deleted by any user, including Super Admins
BR-08	A workflow execution in Paused state CANNOT be deleted; it must be approved, rejected, cancelled, or timed out first
BR-09	AI model credentials provided by organizations MUST be stored encrypted and NEVER returned in plaintext
BR-10	A workflow CANNOT be re-triggered while a previous execution with the same inputs is in progress (idempotency guard) unless explicitly configured to allow concurrent executions
BR-11	An organization owner CANNOT be removed from the organization without transferring ownership first
BR-12	Test executions MUST NOT affect production tool endpoints, production databases, or production credentials unless explicitly configured
BR-13	Deleted documents MUST be de-indexed from the vector store before the deletion is confirmed
BR-14	A user's session MUST be immediately invalidated upon account deactivation or removal from organization
BR-15	Evaluation dataset executions MUST NOT write to production data stores

16. SECURITY REQUIREMENTS

(See Section 14.1 and Section 13.9 for detailed requirements)

Additional security requirements:

ID	Requirement
SEC-10	Privileged actions (e.g., deleting an organization, adding admin) must require re-authentication
SEC-11	The platform must implement security headers (CSP, HSTS, X-Frame-Options, etc.)
SEC-12	All file uploads must be scanned and validated before ingestion
SEC-13	The platform must log all authentication events (success, failure, logout)
SEC-14	Security incidents must trigger automated alerts to platform administrators
SEC-15	External API credentials must be tested for validity before being stored

17. PRIVACY REQUIREMENTS

(See Section 14.2 for detailed requirements)

Additional privacy requirements:

ID	Requirement
PRIV-06	AI agent prompts and outputs may contain user data; this must be subject to data retention and deletion policies
PRIV-07	The platform must have a configurable data masking layer for AI inputs/outputs
PRIV-08	Organizations must be informed of any sub-processors used by the platform

18. AUDIT REQUIREMENTS

(See Section 13.9.4 for detailed requirements)

Audit event categories that must be covered:

- Authentication events (login, logout, failed login, MFA)
 - Identity events (password change, email change, account creation, deletion)
 - Organization events (creation, settings change, member addition/removal)
 - Agent events (create, update, publish, delete, execute)
 - Workflow events (create, publish, execute, cancel, delete)
 - Knowledge events (upload, index, delete, access)
 - Tool events (register, configure, execute, delete)
 - Approval events (request, approve, reject, escalate, delegate)
 - Policy events (create, update, delete)
 - API key events (create, use, revoke)
 - Admin events (role assignment, settings change)
 - Security events (suspicious activity, rate limit hit, validation failure)
-

19. RELIABILITY & FAULT TOLERANCE REQUIREMENTS

ID	Requirement
FT-01	Workflow executions must be persisted to durable storage before being dispatched for processing
FT-02	If the execution engine restarts, in-flight workflow runs must be recoverable and continue from the last successfully committed step
FT-03	External tool calls must have configurable timeouts to prevent indefinite blocking
FT-04	AI model API calls must have configurable timeouts and retry strategies
FT-05	The platform must implement circuit breaker patterns for external service calls
FT-06	Knowledge ingestion failures must be retried automatically with exponential backoff
FT-07	Webhook delivery failures must be retried automatically with exponential backoff
FT-08	Email delivery failures for critical notifications (approvals, failures) must be retried

20. SCALABILITY & PERFORMANCE REQUIREMENTS

(See Section 14.4 and 14.5)

Additional requirements:

ID	Requirement
SCAL-05	The platform must support horizontal scaling of the workflow execution engine using a queue-based architecture
SCAL-06	Knowledge indexing must support large collections (>10,000 documents per organization)
SCAL-07	The audit log system must be designed to handle high write volume without impacting workflow execution performance

21. DATA LIFECYCLE REQUIREMENTS

ID	Requirement
DL-01	Organizations can configure data retention policies for: execution logs, audit logs, AI traces, and evaluation results
DL-02	Data that exceeds the retention period must be automatically purged
DL-03	Data purging must not break the referential integrity of active resources
DL-04	Organizations can export their data (agents, workflows, knowledge metadata) before deletion
DL-05	Platform-initiated data purge events must be logged as audit events
DL-06	Deleted organization data must be fully purged within 30 days of organization deletion

22. AI MODEL MANAGEMENT REQUIREMENTS

ID	Requirement
AI-01	Organizations can configure multiple AI model providers in their organization settings
AI-02	Each AI model provider configuration must store: provider name, API endpoint, and encrypted API key
AI-03	Organizations can define a default model for all agents
AI-04	Individual agents can override the organization default model
AI-05	The platform must support at minimum: OpenAI, Anthropic, and Google Vertex AI model providers (Phase 1)
AI-06	The platform must support provider-agnostic model selection without requiring workflow or agent redesign
AI-07	Organizations can configure model-level policies (e.g., certain models require additional approval)
AI-08	Model availability and health must be monitored; model failures must trigger fallback or failure handling

23. LIFECYCLE DEFINITIONS

23.1 Workflow Lifecycle



State	Description
Draft	Workflow is under construction and cannot be executed
Published	Workflow is active and can be triggered
Disabled	Workflow is paused; no new executions but existing runs complete
Archived	Workflow is retired; read-only; execution history preserved

23.2 Agent Lifecycle



State	Description
Draft	Agent under construction; configuration incomplete or untested
Testing	Agent available for sandboxed test executions only
Published	Agent active and available for workflow use
Deprecated	Agent no longer recommended; existing workflows continue; no new workflows
Archived	Agent retired; read-only; history preserved

23.3 Knowledge Lifecycle

State	Description
Uploading	File transfer in progress
Pending	Upload complete; queued for ingestion
Processing	Ingestion pipeline active (extraction, chunking, embedding)
Indexed	Document successfully indexed and available for retrieval
Failed	Ingestion failed; error details available; retry possible
Deleted	Document removed and de-indexed

23.4 User Lifecycle

State	Description
Registered	Account created; email not yet verified
Active	Verified and has access
Deactivated	Access suspended by admin; data preserved
Deleted	Account and personal data removed per retention policy

23.5 Organization Lifecycle

State	Description
Active	Organization operational; all features available
Suspended	Organization suspended (e.g., non-payment); members cannot log in
Deleted	Organization and all data scheduled for purge per retention policy

24. RISK MANAGEMENT

24.1 Technical Risks

Risk	Probability	Impact	Mitigation
AI model API downtime	Medium	High	Multi-provider support; fallback strategies; graceful error handling
Vector DB performance degradation	Low	High	Performance testing; caching; index optimization
Workflow engine state corruption	Low	Critical	Durable state persistence; transaction logs; recovery procedures
Cross-tenant data leak	Very Low	Critical	Architectural isolation; rigorous testing; security audits
Token cost overruns	Medium	Medium	Budget alerts; hard limits per organization; cost tracking
Prompt injection attacks	Medium	High	Input validation; output filtering; guardrail policies

24.2 Business Risks

Risk	Probability	Impact	Mitigation
AI hallucinations causing incorrect business actions	Medium	High	HITL for high-risk; evaluation framework; groundedness scoring
Organizations misusing autonomous actions	Medium	High	Risk classification; approval policies; audit trails; ToS enforcement
Regulatory changes affecting AI usage	Medium	Medium	Policy-driven architecture allowing rapid configuration changes
Competitor feature parity	Medium	Medium	Depth of governance, observability, and HITL as differentiators

25. COMPLIANCE CONSIDERATIONS

The platform must be designed with the following compliance frameworks in mind:

Framework	Applicability
GDPR	For EU-based organizations and their data subjects
SOC 2 Type II	Enterprise customers will require this certification
ISO 27001	Information security management
CCPA	California-based organizations and their users
HIPAA	If healthcare organizations use the platform (Phase 3, explicit scope)

Note: Achieving these certifications is not a requirement for MVP. The platform must be designed to be compatible with these requirements from the outset.

26. REAL-WORLD BUSINESS SCENARIOS

The following scenarios illustrate how the platform supports different business use cases. These are reference scenarios, not prescriptive implementations.

Scenario 1: Customer Support Automation

Organization Type: SaaS company

Workflow: Incoming support ticket → AI agent reads ticket and knowledge base → Drafts response → Human approval for complex cases → Sends response → Logs outcome

Key Features Used: Webhook trigger, AI Agent Node, RAG Node, Human Approval Node, Tool Node (email/CRM)

Scenario 2: Sales Lead Qualification

Organization Type: Product-based company

Workflow: New CRM lead event → AI agent qualifies lead (scoring, intent, ICP fit) → Routes to appropriate sales rep or auto-nurture campaign

Key Features Used: Event/Webhook trigger, AI Agent Node, RAG Node (product knowledge), Condition Node, Tool Node (CRM update, email)

Scenario 3: Client Onboarding

Organization Type: Service-based consulting firm

Workflow: New client contract signed → AI agent generates onboarding document package → Human approval → Documents sent to client → Kickoff tasks created

Key Features Used: Manual/API trigger, AI Agent Node, RAG Node (template knowledge), Human Approval Node, Tool Node (document creation, project management tool)

Scenario 4: Internal Knowledge Assistant (Workflow-Backed)

Organization Type: Any

Workflow: User submits a question → AI agent retrieves relevant documents → Generates grounded answer → Returns answer with citations

Key Features Used: Manual trigger, AI Agent Node, RAG Node, Notification Node

Scenario 5: IT Service Ticket Automation

Organization Type: Any

Workflow: IT ticket created → AI agent classifies and prioritizes ticket → Resolves standard issues autonomously → Escalates to human for non-standard issues

Key Features Used: Webhook trigger, AI Agent Node, RAG Node (IT runbooks), Condition Node, Human Approval Node, Tool Node (ticketing system)

Scenario 6: Document Processing

Organization Type: Professional services, finance

Workflow: Document uploaded → AI agent extracts key data fields → Validates against rules → Outputs structured data → Flags exceptions for review

Key Features Used: Event trigger, AI Agent Node, Condition Node, Human Approval Node (for exceptions), Tool Node (database write)

Scenario 7: Compliance Report Generation

Organization Type: Any

Workflow: Scheduled trigger → AI agent pulls data from connected systems → Analyzes against compliance rules → Generates report → Routes to compliance officer for approval → Distributes

Key Features Used: Scheduled trigger, AI Agent Node, RAG Node, Tool Node (data pull), Human Approval Node, Notification Node

Scenario 8: Email Classification and Routing

Organization Type: Any

Workflow: Incoming email → AI agent classifies intent → Routes to correct department workflow or auto-responds → Logs event

Key Features Used: Webhook/email trigger, AI Agent Node, Condition Node, Tool Node (email/CRM)

Scenario 9: Contract Review Workflow

Organization Type: Legal, HR, sales

Workflow: Contract document uploaded → AI agent reviews against policy knowledge base → Flags high-risk clauses → Routes to legal team for review → Documents decision

Key Features Used: Manual trigger, AI Agent Node, RAG Node (legal policy knowledge), Human Approval Node, Tool Node

Scenario 10: Recurring Operations Workflow

Organization Type: Operations, finance

Workflow: Scheduled trigger → AI agent pulls metrics from connected tools → Generates operations summary → Posts to team channel → Flags anomalies for attention

Key Features Used: Scheduled trigger, AI Agent Node, Tool Node (data pull, messaging), Condition Node, Notification Node

27. FUTURE EXTENSIBILITY

The platform must be designed to accommodate the following future capabilities without requiring a redesign:

Future Capability	Consideration
Multi-modal AI (vision, audio)	Agent and node type system must be extensible
Real-time streaming responses	Execution model must support streaming output
Multi-agent collaboration	Agent orchestration must support agent-to-agent communication
Industry-specific templates	Workflow and agent template marketplace concept
White-label/embedded product	Multi-tenant with white-label brand settings
Mobile-responsive experience	Web-first; mobile-responsive from launch
Advanced fine-tuning	Model management must allow custom model endpoints
Third-party plugin ecosystem	Tool registry must support external plugin registration
Data residency controls	Data storage must be geo-configurable

28. PRODUCT ROADMAP

MVP Scope

Goal: Deliver the smallest genuinely useful production-quality product that demonstrates the full core value chain:

User → Organization → Knowledge → Agent → Workflow → Tool → Approval → Execution → Monitoring

MVP Inclusions:

- Email/password authentication with email verification and MFA (TOTP)
- Organization creation and member management
- Role-based access control (standard roles)
- AI Agent creation and configuration (Draft → Published lifecycle)
- Knowledge collection creation and document upload with RAG
- Built-in tools: HTTP/REST, Email, Data Transform
- Visual workflow builder with core node types (Trigger, Agent, RAG, Tool, Condition, Approval, Notification, End)
- Manual and scheduled triggers
- Human-in-the-loop: approval queue, approve/reject
- Workflow execution engine with status tracking, retries, and failure handling
- Basic execution monitoring: run history, step logs, error tracking
- Basic observability: token usage, cost, latency per run
- Basic AI evaluation: task completion and groundedness scoring
- Basic analytics: workflow success rate, execution volume, cost
- Audit logging (core events)
- Organization administration: user management, AI model settings, basic policies
- In-app and email notifications

MVP Out of List (Explicitly deferred):

- API/Developer platform (REST API, API keys, webhooks)
 - Agent versioning and rollback
 - Workflow versioning
 - Sub-workflow nodes
 - Loop nodes
 - OAuth connectors for external systems
 - Advanced AI evaluation (hallucination detection, custom criteria)
 - Advanced analytics (exports, custom reports)
 - SSO (SAML/OAuth)
 - Advanced governance (data classification, custom AI policies)
 - Workspace-level scoping
-

Phase 2 Scope

Goal: Expand the platform into a full-featured commercial product with developer access and advanced capabilities.

Phase 2 Inclusions:

- Developer API (REST API, API keys, scoped access)
 - Outbound webhooks for workflow events
 - API and Webhook triggers for workflows
 - Sub-workflow nodes
 - Loop nodes
 - Agent versioning and rollback
 - Workflow versioning
 - OAuth-based integration connectors (Phase 2 connectors: CRM, ticketing systems, calendar)
 - Hybrid knowledge retrieval (semantic + keyword)
 - Advanced AI evaluation (hallucination detection, custom evaluation criteria)
 - Approval delegation and escalation
 - Workspace-level scoping within organizations
 - Advanced analytics (CSV export, custom date ranges, per-agent/per-workflow drill-down)
 - Agent dry-run mode
 - Web search built-in tool
 - Agent-to-agent handoff (multi-agent)
 - Advanced guardrail policies (PII detection, custom prohibited topics)
 - Cost budget enforcement (hard limits per organization)
 - IP allowlisting for API access
-

Phase 3 Scope

Goal: Enterprise-grade capabilities for regulated industries and large-scale deployments.

Phase 3 Inclusions:

- SSO via SAML 2.0 and enterprise OAuth
- HIPAA-compliance features (explicit scope if healthcare vertical targeted)
- SOC 2 Type II certification support features
- Data residency controls (geographic data isolation)
- Custom audit log export integrations (SIEM integration)
- Fine-grained role customization (custom role definition beyond standard roles)
- Enterprise SLA features (dedicated infrastructure options)
- White-label capabilities
- Advanced multi-agent collaboration (agent teams, shared memory)
- Evaluation regression testing framework
- Third-party plugin/tool ecosystem

- Programmatic agent creation via API
 - Bulk workflow operations
 - Full GDPR data subject request automation
 - Platform API for embedding in third-party products
-

Future Vision

The long-term vision for AI Workforce includes:

- An AI Workforce Marketplace where organizations can share or purchase pre-built agent templates, workflow templates, and integration packs
 - An embedded AI Workforce module that can be white-labeled into third-party products
 - Multi-modal agent capabilities (processing images, audio, and structured data alongside text)
 - AI Workforce Analytics across industry benchmarks (anonymized, opt-in)
 - Real-time collaborative workflow building for teams
-

29. ACCEPTANCE CRITERIA FRAMEWORK

All features must meet the following acceptance criteria dimensions before being considered complete:

1. **Functional Correctness:** The feature behaves as specified in the requirements
 2. **Role/Permission Validation:** Access control is correctly enforced for all roles
 3. **Multi-Tenant Isolation:** Data is correctly scoped and isolated per organization
 4. **Error Handling:** Failure cases are gracefully handled with appropriate error responses
 5. **Audit Trail:** All significant actions produce the required audit events
 6. **Performance:** The feature meets defined performance benchmarks
 7. **Security:** The feature passes security validation (no injection vulnerabilities, proper authorization)
 8. **Observability:** Key events are logged and traceable
-

30. RISKS & MITIGATIONS

Risk	Likelihood	Severity	Mitigation Strategy
AI model provider outage	Medium	High	Multi-provider architecture; fallback model selection
Prompt injection compromise	Medium	High	Input validation; sandboxing; output filtering; audit
Cross-tenant data leak	Low	Critical	Architectural enforcement; automated security testing
Unexpected AI cost spike	High	Medium	Budget alerts; hard limits; cost dashboards
Slow knowledge ingestion at scale	Medium	Medium	Async pipeline; scalable processing; progress feedback
Approval queue neglect causing workflow deadlocks	Medium	Medium	Configurable timeouts; escalation chains; auto-reject
Regulatory changes impacting AI usage	Medium	High	Policy-driven architecture; regular compliance review
Data loss during execution engine failure	Low	Critical	Durable message queues; idempotent steps; recovery procedures

31. OPEN QUESTIONS & DECISIONS REQUIRED

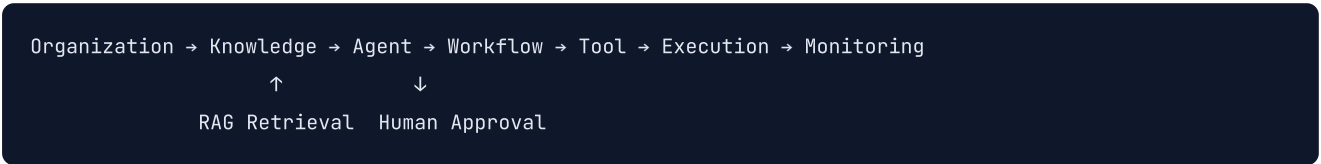
The following questions must be resolved before architectural design begins:

ID	Question	Owner	Priority
OQ-01	Will the platform offer managed AI model hosting, or will all organizations bring their own API keys?	Product/Business	Critical
OQ-02	Which specific AI model providers will be supported at MVP launch?	Product/Engineering	Critical
OQ-03	What is the exact multi-tenant data isolation strategy: shared schema with row-level security, schema-per-tenant, or database-per-tenant?	Engineering	Critical
OQ-04	What is the vector database selection strategy?	Engineering	Critical
OQ-05	Will the platform offer a "platform-hosted" credential option where we provide default AI model API keys (metered billing) or require all organizations to use their own?	Product/Business	High
OQ-06	What are the specific tier limits (workflow runs, tokens, storage) for each subscription plan?	Product/Business	High
OQ-07	What is the exact list of built-in connectors at Phase 2 launch?	Product	High
OQ-08	Will evaluation be AI-assisted (LLM-as-judge) or rule-based only at MVP?	Engineering	High
OQ-09	What data masking approach will be used for PII in AI traces?	Engineering/Legal	Medium
OQ-10	Should the MVP include real-time streaming of AI agent reasoning, or is final output sufficient?	Product	Medium
OQ-11	What is the retention policy for AI execution traces (default and maximum)?	Product/Legal	Medium
OQ-12	Will the approval queue support mobile-responsive access at MVP?	Product	Low
OQ-13	Is a free trial tier (time-limited or execution-limited) required at MVP launch?	Business	High

32. FINAL PRODUCT REQUIREMENT SUMMARY

AI Workforce is a **multi-tenant, enterprise-grade AI workflow automation SaaS platform** designed to allow businesses to deploy an autonomous AI digital workforce governed by security, audit, and human oversight controls.

Core Value Chain



Platform Pillars

Pillar	Purpose
Identity & Access	Secure, role-controlled access for every user action
Agent Management	Create, configure, version, and govern AI agents
Knowledge Management	Upload, index, and retrieve organizational knowledge
Workflow Orchestration	Visually build and execute multi-step AI workflows
Human-in-the-Loop	Enforce human oversight for high-risk AI actions
Observability	Full visibility into every AI action and workflow execution
Governance	Risk classification, guardrails, audit, and policy enforcement
Evaluation	Continuous quality measurement of AI outputs
Analytics	Business-level ROI and automation metrics
Developer Platform	API access for programmatic integration and extensibility

What Makes This Different

- 1. Not a chatbot** — It is a workflow automation operating system with AI reasoning embedded
- 2. Not a RAG demo** — Knowledge management is one pillar of a complete platform
- 3. Not a single-use tool** — The platform is horizontal and supports any knowledge-intensive workflow
- 4. Governance by design** — Security, audit, HITL, and guardrails are not features; they are architectural foundations
- 5. Observable by default** — Every AI action is traced, logged, and measurable
- 6. Production-grade** — Built for reliability, scalability, fault tolerance, and enterprise adoption from day one

End of Business Requirements Document

Document Version: 1.0.0 | Status: Draft Next Action: Stakeholder Review and Sign-off before proceeding to Architecture Phase